

Rulebook Drift Monitor

Meeting handout & build plan · CDIR Global 'Agentic Regulator' Hackathon (team dev overview)

Synthetic-data disclaimer

All data in this prototype is synthetic and illustrative. The DS rulebook (DS-01..DS-40) is a paraphrased demonstration schema based on FATF (2020) red-flag indicators and FIC Directive 9 / FIC Act obligations; rule text is not verbatim from source and must be verified against the source documents before any production use. The 35 fraud typologies and their transaction fixtures (TYP-001..TYP-035) are fabricated examples derived from public reporting themes for demonstration only and contain no personal data. Nothing in this prototype reflects live customer, transaction, or supervisory data.

1 What the project is

An **agentic system** for the FIC and FSCA that continuously **stress-tests the crypto AML/KYC rulebook** against both documented and self-generated fraud typologies, and hands each **verified detection gap** to a named analyst as a **drafted candidate red flag**. It sits in the market-monitoring and horizon-scanning stage of the regulatory life cycle and hands into policy development, with a human approval gate on that boundary.

Problem: AI-driven crypto fraud is exploding (TRM: ~500% rise in AI-enabled crypto scams in 2025), but SA's CASP rulebook relies on static, rule-based detection that predates generative AI (the FATF red-flag baseline is from 2020). Current flow is reactive and takes months: scam → victims → police → guidance → manual rule updates. Our system compresses that lag to days.

2 What you're seeing in the demo

The run produces a **Ranked Gap Report**. Each finding shows:

- **Typology** — the fraud pattern being stress-tested (e.g. pig-butcher/romance scam, AI-cloned-voice phishing, synthetic identity).
- **MITRE ATLAS** — the adversarial-machine-learning framework mapping of that pattern's techniques.
- **Fires rules (green +)** — existing indicators that DID trigger; **Evades rules (red -)** — indicators that should have caught this pattern but DID NOT (the stale/gap rules).
- **Drafted candidate red flag** — a single-sentence screening indicator a compliance team should adopt, generated by the LLM (or deterministic template).
- **Evidential basis** — where the typology came from (documented source, or novel self-generated).

Two things prove the **guardrails** work:

- **Critic rejection:** a 'Rejected by critic' section shows an implausible/speculative novel generation was **discarded** (no agent marks its own work).
- **Human approval gate:** the report ends waiting for a named analyst to **accept / amend / reject** each finding — nothing is promoted into guidance automatically.

The run is backed by a full **audit trail** (auditability & traceability): every step is logged with its node, message, rulebook version and run id.

6 Live demo output (from an actual run)

Run id: **run-b39c5c27** — exactly what a triggered run produces.

#	1
Finding	Investment/Ponzi crypto fraud (TYP-001)
Fires (rule DID trigger)	DS-01, DS-02, DS-04, DS-28, DS-31, DS-37
Evades (rule IS stale/gap)	DS-06, DS-09, DS-12, DS-17, DS-26, DS-39
MITRE ATLAS	TA0001 Reconnaissance, T1552 Phishing, AML.T1566 Social Engineering
Evidence	documented typology
Drafted candidate red flag	"Customer engages in activity consistent with 'Investment/Ponzi crypto fraud' that is not adequately detected"

#	2
Finding	Pig-butcherer / romance scam (AI persona) (TYP-002)
Fires (rule DID trigger)	DS-08, DS-09, DS-12, DS-15, DS-40
Evades (rule IS stale/gap)	DS-04, DS-06, DS-26, DS-39
MITRE ATLAS	T1552 Phishing, T1566 Social Engineering, AML.T0016.002 Generative AI
Evidence	documented typology
Drafted candidate red flag	"A candidate's use of deepfake personas, deepfake media, synthetic identities, or social engineering tactics"

#	3
Finding	Mule / money-mule pass-through network (TYP-003)
Fires (rule DID trigger)	DS-04, DS-05, DS-23, DS-25, DS-26
Evades (rule IS stale/gap)	DS-12, DS-15, DS-40
MITRE ATLAS	TA0006 Impact, TA0007 Evasion, T1071 Layer
Evidence	documented typology
Drafted candidate red flag	"A financial institution's customer is opening a new, unregistered, high-value account in a foreign jurisdiction"

#	4
Finding	AI-cloned voice / phishing of account access (TYP-004)
Fires (rule DID trigger)	DS-06, DS-13, DS-24
Evades (rule IS stale/gap)	DS-12
MITRE ATLAS	T1552 Phishing, TA0001 Reconnaissance, T1566 Social Engineering
Evidence	documented typology
Drafted candidate red flag	"A Customer is attempting to initiate a transaction using a voice clone or phishing call, or is using a cloned voice"

#	5
Finding	Synthetic identity new-customer fraud (TYP-005)
Fires (rule DID trigger)	DS-01, DS-09, DS-28, DS-39
Evades (rule IS stale/gap)	DS-06, DS-07, DS-23
MITRE ATLAS	TA0005 ML Development, TA0007 Evasion, T1592 Gather Victim Org Info
Evidence	documented typology
Drafted candidate red flag	"Compliance team should watch for suspicious transactions involving rapid volume, new account opening"

#	6
Finding	Stablecoin cross-border payment structuring (TYP-006)
Fires (rule DID trigger)	DS-02, DS-03, DS-19, DS-34, DS-37
Evades (rule IS stale/gap)	DS-20, DS-24, DS-38
MITRE ATLAS	TA0007 Evasion, T1071 Layer
Evidence	documented typology
Drafted candidate red flag	"A customer is exhibiting suspicious activity in their account by consistently structuring transactions into sm

#	7
Finding	Romance scam with AI deepfake video persona (TYP-007)
Fires (rule DID trigger)	DS-01, DS-04, DS-09, DS-12, DS-26, DS-37, DS-40
Evades (rule IS stale/gap)	DS-28, DS-39
MITRE ATLAS	AML.T0016.002 Generative AI, T1566 Social Engineering, TA0007 Evasion
Evidence	documented typology
Drafted candidate red flag	"Customer engages in activity consistent with 'Romance scam with AI deepfake video persona' that is not a

#	8
Finding	Deepfake liveness-bypass document fraud (TYP-008)
Fires (rule DID trigger)	DS-01, DS-09, DS-28, DS-39
Evades (rule IS stale/gap)	DS-06, DS-07, DS-08, DS-23, DS-27, DS-29
MITRE ATLAS	TA0005 ML Development, TA0007 Evasion
Evidence	documented typology
Drafted candidate red flag	"A candidate's ability to create, manipulate, or alter digital content in a way that deceives or misleads other

...plus 29 further verified findings (total 37), one per evaluated typology/fixture — the same schema as above. Full ranked report is in the demo.

Rejected by critic (guardrail — not promoted):

- prompt-poisoned global settlement reroute (GEN-NOVEL) — rejected by critic: speculative self-generated path with no supporting published evidence or STR material

How to read it: 'Evades rules' are the existing FIC/Directive-9/FATF indicators that should have caught that fraud pattern but did not — the detection gap. 'Fires rules' are the ones that did trigger. The drafted candidate red flag is the proposed new screening indicator a compliance team could adopt. Findings marked GEN-NOVEL are self-generated novel evasion paths (subject to expert plausibility review); the critic rejects unsupported ones. All rules are alphanumeric DS-01..DS-40 identifiers from the Drift Sentinel schema.

7 Prediction layer (forecast analytics)

A transparent **drift-index overlay** is computed in real time from the deterministic library scan (35 typologies × 40 rules = 219 relevant pairs). Today: **coverage 48%**, **drift index 55/100** (Elevated, projected to 75.8 in 6 review cycles at current AI-evasion pressure of 66%). Most at-risk next period: TYP-021 (ESports skin-gambling 0.77), TYP-001 (Investment/Ponzi crypt 0.72), TYP-002 (Pig-butcherer / roman 0.72).

Projection source: **measured history** — fitted to 8 recorded runs/checkpoints of the real drift-index series (cadence ~21.0d) with a widening confidence band. With a roll-one-out backtest: MAD 1.01, MAPE 1.8%, bias 1.013 (n=6)

Methodology: Drift Index = coverage deficit (65 weight) + AI-evasion pressure (20) + severity floor (10) + AI-novelty concentration (5). With 3+ recorded runs/checkpoints the projection is a damped trend fitted to the MEASURED drift series (least-squares slope combined with the measured AI-evasion rate) with a confidence band and hold-out backtest; below that it falls back to a clearly-labelled damped exponential smoothing baseline. It is a **decision-support model**, not a calibrated statistical forecast — every number is reproducible from the library scan and recorded history.

Sample finding (abridged):

Finding	Stablecoin cross-border payment structuring (TYP-006)
Fires	DS-02, DS-03, DS-19, DS-34, DS-37
Evades	DS-24, DS-38
ATLAS	TA0007 Evasion · T1071 Layer
Drafted flag	"Structuring stablecoin cross-border payments to exceed thresholds without disclosing originator/beneficiary info m

Note: 'Fires' and 'Evades' can list the same rule when a rule fires on one fixture of the typology but is evaded by another adversarial fixture. The key is which indicators the typology overall defeats.

3 Architecture

Agent	Responsibility
Orchestrator	Decomposes the run, routes tasks, runs the two arms in parallel, produces the report.
Retrieval agent	Grounds the system: surfaces relevant rules per typology, selects threat intel, drafts candidate red flags
Simulation agent	Two modes: reconciliation (documented typologies) + generation (novel evasion paths).
Critic agent	Independent verifier; re-runs every claim deterministically, discards non-reproducible / implausible findings
Rule-evaluation engine	Deterministic core that steps a synthetic transaction through each rule and records fire/evade — reproducible

4 How to run the project

Prerequisites: Python 3.9+, and Ollama for the local free LLM (optional).

One-time setup

```
brew install ollama
ollama pull llama3.2:1b
pip install langgraph langchain langchain-openai langchain-community flask requests
```

Run

```
cd /Users/kyria/Documents/benji_work/Hackathon/rulebook_drift_monitor
```

Option A — CLI ranked gap report:

```
python3 -m demo.run
python3 -m demo.run --audit # include audit trail
python3 -m demo.run --json # raw state
```

Option B — Web demo (open <http://127.0.0.1:5000/>):

```
python3 -m demo.web
```

If Ollama is off, the system degrades gracefully (deterministic templates) so it never crashes. Make sure **ollama serve** is running once; if the port is already in use, it's already up.

5 8-day build plan (deadline 8 Sep, 23:59 AOE)

Submission = presentation deck + ≤3-min video demo + testable prototype (NayaOne GitLab). Demo Days 15–16 Sep, Judging 17 Sep. Status as of today (31 Aug): core end-to-end slice DONE.

When	What to do
Day 1 · Tue 1 Sep (kick-off)	Finalise roles; expand rulebook to 30–40 rules; init NayaOne GitLab repo + requirements.txt; shake
Day 2 · Wed 2 Sep (knowledge clinic)	Finalise and typologies to 12–15 with richer fixtures; add real vector store (ChromaDB/FAISS) for RAG; s
Day 3 · Thu 3 Sep (knowledge clinic)	Strengthen critic: back-testing re-discovers seeded gaps + plausibility scoring; evaluation harness (r
Day 4 · Fri 4 Sep	Integrate vector store + ATLAS mapper + evaluation into workflow; harden audit trail + human-in-the
Day 5 · Mon 7 Sep (pitch clinic)	Build 4-min presentation deck (NayaOne template); refine demo/video script; book ≥1 mentor sessio
Day 6 · Tue 8 Sep (pitch clinic + DEADLINE)	Finalise video (end-to-end, guardrails live, downloadable); finalise prototype + README; subm
Buffer (9–16 Sep)	Polish visuals; rehearse 4-min pitch to the clock; load-test demo reliability; prep Q&A answers.

Task ownership (suggested; combine roles for a smaller team)

- **Dev A (core):** engine, agents, workflow, integration lead.
- **Dev B (data):** rulebook + typology expansion, fixtures, vector store.
- **Dev C (eval):** MITRE ATLAS mapper, evaluation harness, back-testing.
- **Dev D (product):** web app polish, video demo, NayaOne deployment.
- **Pitch/Product:** presentation deck, demo narrative, guardrails story.

Guardrails (all mandatory, all demonstrated in the demo):

- **Human-in-the-loop** — analyst accepts/amends/rejects every red flag.
- **Auditability & traceability** — full audit trail linking each gap to evidence, rules, ATLAS and analyst.
- **Safety & governance** — independent critic discards implausible findings; generation held at indicator level.
- **Cyber risk management** — permissioned tool access, confined corpora, public/synthetic data only.

Deliberately NOT built: live blockchain analytics, entity supervision, model fine-tuning. Post-hackathon: extra corpora, FIC STR intake integration, supervised pilot.